

Multi-Perspective Business Analysis - AdventureWorks Cycles

Sales Analytics Perspective

Prepared by: Faith Aipoh
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Perspective & Focus Area

I chose the Sales Analytics perspective to support the Sales Director and executive team in understanding revenue drivers, product performance, territory effectiveness, customer retention, and seasonal patterns. The analysis covers the company's global sales of bicycles, components, clothing, and accessories using the AdventureWorks dataset (2011-2014).

The dashboard comprises five pages:

1. Executive KPI dashboard
2. Product Performance
3. Customer Insights
4. Geographic & Seasonal Trends
5. Actionable Recommendation

This report summarizes the key findings, assumptions, methodology, and recommended business actions.

Key Assumptions

- **Revenue calculation** = $\text{OrderQty} * \text{UnitPrice}$. No returns or discounts were subtracted because dataset does not contain a return table.
- **Customer definition** - Unique CustomerID in SalesOrderHeader. A repeat customer is any customer with more than one order.
- **Time period** - Data spans calendar years 2011 to 2014. Note: 2011 records start in May; 2014 records stop in June. Seasonality comparisons treat these as partial years where noted.
- **Geography** - Territory names and country codes are as provided.
- **Online vs In-Store** - OnlineOrderFlag = 1 means online order; 0 means in-store.

Methodology & Derivation

Data preparation

- Loaded seven tables into Power BI: SalesOrderHeader, SalesOrderDetail, SalesTerritory, Customer, Product, ProductCategory, ProductSubCategory.
- Created relationships:
 - ✧ SalesOrderDetail[SalesOrderID] 'n SalesOrderHeader[SalesOrderID]
 - ✧ SalesOrderHeader[CustomerID] 'n Customer[CustomerID]
 - ✧ SalesOrderHeader[TerritoryID] 'n SalesTerritory[TerritoryID]
 - ✧ SalesOrderDetail[ProductID] 'n Product[ProductID]
 - ✧ Product[ProductSubCategoryID] 'n ProductSubCategory[ProductSubCategoryID]
 - ✧ ProductSubCategory[ProductCategoryID] 'n ProductCategory[ProductCategoryID]

DAX measures created

- Total Revenue = $\text{SUMX}('Sales\ SalesOrderDetail', 'Sales\ SalesOrderDetail'[\text{OrderQty}] * 'Sales\ SalesOrderDetail'[\text{UnitPrice}])$
- Total Orders = $\text{DISTINCTCOUNT}('Sales\ SalesOrderHeader'[\text{SalesOrderID}])$
- Avg Order Value = $\text{DIVIDE}([\text{Total Revenue}], [\text{Total Orders}])$
- Total Quantity = $\text{SUM}('Sales\ SalesOrderDetail'[\text{OrderQty}])$
- Total Customer = $\text{DISTINCTCOUNT}('Sales\ SalesOrderHeader'[\text{CustomerID}])$
- Avg Revenue per Customer = $\text{DIVIDE}([\text{Total Revenue}], [\text{Total Customer}], 0)$
- Repeat Customer = $\text{COUNTROWS}(\text{FILTER}(\text{VALUES}('Sales\ SalesOrderHeader'[\text{CustomerID}]), \text{CALCULATE}(\text{DISTINCTCOUNT}('Sales\ SalesOrderHeader'[\text{SalesOrderID}])) > 1))$
- Repeat Rate % = $\text{DIVIDE}([\text{Repeat Customer}], [\text{Total Customer}], 0)$

Calculated Columns

- OrderYear from OrderDate to enable seasonal grouping.
- CustomerOrderSequence and Customer Type (New / Returning) per order.
- Order Type (Online / In-Store) to see which sales channel customer prefer.

Table created

I created an OrderFrequency table by adding CustomerID from SalesOrderHeader, then I created two calculated columns Order Count and OrderBucket. Order Count is the distinct count of SalesOrderID. OrderBucket is a column that categories Order Count into three groups (e.g., Order Count = 1 -> 1 order, Order Count = 2 -> 2 order, Order Count >= 3 -> 3+ order)

Visualisation choices

- KPIs cards for high-level metrics.
- Bar charts and treemaps for product hierarchy and territory revenue.

- Line charts with year-as-legend for seasonal trends.
- Waterfall chart for month-over-month changes.
- Donut charts for order and customer types.
- Scatter chart for product prices and quantity sold.

All visuals use consistent company colors (dark blue, orange accent) and slicers for year, territory, category, customer type, and order type.

Insights & Recommendations

Product Performance

- **Concentration risk** - Top 3 products account for **86.2% of total revenue**. Road bikes lead (\$43.9M), touring bikes lag (\$14.55M).
- **Category revenue** - Bikes = \$95M, Accessories only \$1.2M. Clothing and Accessories sell high unit volumes (73,670 and 61,932 unit respectively) but low revenue.
- **Recommendations**
 - ✧ Ensure top-3 SKUs never out of stock.
 - ✧ Investigate touring bike pricing/marketing; consider bundling with accessories.
 - ✧ Test small price increases on high-volume, low value clothing/accessories.

Customer Retention

- **Total customers** = 19,119; **repeat customers** = 7,470 (39%).
- Repeat customers generate **71.4% of revenue**. Only 28.6% of customers buy twice, and only 10.5% buy three or more times.
- Average revenue per customer = \$5,773.
- **Recommendations**
 - ✧ Launch a post-purchase email sequence targeting first-time buyers.
 - ✧ Create a tiered loyalty program (2nd purchase = 5% off, 3rd = 10% off).
 - ✧ Identify the top 10% of repeat customers for VIP treatment.

Territory UnderPerformance

- **Germany** is the most underperforming territory: \$4.95M revenue (just 4.48% of total) from 2,623 orders.
- This suggests low average order value or weak local market penetration.
- **Recommendations**
 - ✧ Analyse Germany's average order value vs company average.
 - ✧ Test free shipping thresholds or product bundling.
 - ✧ Run a pilot digital campaign featuring top-selling road bikes.

Seasonality

- **Highest revenue months** - March (\$13.6M, 12.4%), May (\$12.3M, 11.2%), October (\$11.9M, 10.8%).
- **Lowest month** - February (\$5.1M, 4.7%).
- **Product category seasonality**
 - ✧ Bikes peak in March and October.
 - ✧ Components peak May-July.
 - ✧ Accessories and clothing peak in March-May.
- **Channel seasonality** - Online orders peak in May and November; in-store purchase peak in March and October.
- **Recommendations**
 - ✧ Build inventory for Bikes and Components by April and September.
 - ✧ Run a "Winter Clearance" in February to lift the slow month.
 - ✧ Align online ad spend with April (before May peak) and October (before November online peak).

Conclusion

AdventureWorks Cycles' revenue is heavily concentrated in a few products and the Bikes category. Repeat customers are the company's most valuable asset, yet retention beyond a second purchase is low. Germany represents a clear growth opportunity if local barriers can be identified and addressed. Seasonality is strong and predictable, the company can improve profitability by aligning inventory, promotions, and channel-specific campaigns with the March-May and October peaks, while mitigating the February trough.

Next steps for leadership:

1. Approve a Q4 pilot campaign for Germany.
2. Set a target to increase repeat purchase rate from 39.1% to 45% within 12 months.
3. Review touring bikes strategy and accessory pricing.